

Chapter 4

Advanced Diffusion Concepts

Exam Study Guide, Worked Concepts, Sample Questions, and

Prepared from	uploaded course materials
Main source	Chapter_4_Adv_Diffusion(1).ipynb
Exam source	Chapter_4_Summary_and_Sample_Quesions.pdf
Practice source	Chapter_4_Flashcards.pdf

How to use it

Use this review after the basic DDPM chapter. Focus on ELBO-to-MSE, score fields, reverse drift plus stochasticity, classifier guidance, classifier-free guidance, and FID.

Study order: First read the concept sections, then solve the five sample exam questions, then test yourself with the flashcards.

Chapter 4 Advanced Diffusion Concepts

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- 1. Quick exam map
- 2. Why MSE appears in diffusion training
- 3. Score perspective
- 4. Reverse diffusion as drift plus stochasticity
- 5. Classifier guidance
- 6. Classifier-free guidance
- 7. Noise-aware classifiers and tradeoffs
- 8. FID and evaluation
- 9. Formula cheat sheet and symbol key
- 10. Homework and code prompts
- 11. Exam-style questions
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Most useful mental shortcut
Advanced diffusion says the denoiser is learning time-dependent probability geometry, not merely removing noise.

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1. Quick exam map

Advanced diffusion explains what the denoiser means geometrically and how conditional generation is controlled.

Noisy sample x_t

current point in noisy distribution Score $\nabla_x \log p_t(x)$ direction of higher probability

Reverse drift

moves sample toward data

Guidance

adds condition-specific push

Evaluation

checks quality and diversity Figure: the main method pipeline for this chapter.

Exam topic	What you must be able to say
ELBO to MSE	DDPM noise MSE emerges from a variational objective under Gaussian assumptions.
Score	$\nabla_x \log p_t(x)$ points toward higher density at a given noise level.

Exam topic	What you must be able to say
Reverse dynamics	Sampling combines deterministic drift and stochastic noise.
Classifier guidance	Use a noise-aware classifier gradient to bias samples toward y .
CFG	Train conditional and unconditional branches, then combine predictions.
FID	Compare real/generated feature means and covariances, but treat it as a proxy.

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2. Zero-prior explanation: understand the

idea before the equations This section assumes you have not read the lecture yet. The goal is to build intuition first, then connect the intuition to the equations and exam language.

Why MSE appears

The simple noise-prediction loss comes from the probabilistic DDPM structure. With Gaussian forward conditionals and fixed/appropriate reverse variance, the intermediate KL terms reduce to errors in the reverse mean, which can be represented by noise prediction.

Score perspective

The score $\nabla_x \log p_t(x)$ points in the direction where the noisy-data distribution becomes more likely. For Gaussian-corrupted data, predicting noise is closely tied to predicting this score.

Reverse drift plus stochasticity

The deterministic drift moves samples toward high-probability data regions. The stochastic term preserves sampling diversity and prevents the process from becoming pure mode-seeking gradient ascent.

Memory line
Guidance is a knob, not a free improvement.

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3. Core concepts and worked explanations

Classifier guidance

Bayes rule shows that a conditional score can be decomposed into an unconditional score plus a classifier gradient. The diffusion model provides realism; the classifier provides condition alignment.

Classifier-free guidance

CFG trains one model with the condition sometimes present and sometimes dropped. At sampling time, conditional and unconditional predictions are combined to strengthen condition adherence.

Evaluation and FID

FID compares real and generated feature distributions using means and covariances. It is useful but not absolute; it can miss prompt correctness, diversity problems, and domain mismatch.

4. Method explanation, worked example,

and application

Step	Method explanation
Learn score-related	field Noise prediction can be interpreted as learning a denoising/score direction.
Add condition	Use classifier guidance or CFG to bias generation toward y or text.
Tune guidance scale	Increase adherence carefully because diversity may fall.
Evaluate critically	Use FID plus visual and semantic checks.

Worked example
Example: in text-to-image generation, increasing CFG scale usually makes images follow the prompt more strongly, but too much guidance can cause repeated patterns, unnatural textures, or reduced diversity.

Applications and usage
Usage: these ideas power class-conditional generation, text-to-image systems, image editing, controllable synthesis, and quality-diversity tuning.

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5. Formula / definition cheat sheet

Read formulas as compact sentences. Do not memorize symbols blindly; use the symbol key table after the formula table to understand what each part represents.

Formula	Interpretation
$s_t(x)=\nabla x \log p_t(x)$ Score at noise level t .	$s_t(x_t)\approx -\epsilon\theta(x_t,t)/\sqrt{(1-\alpha_t)}$ Noise prediction to score identity.
$\nabla x \log p(x y)=\nabla x \log p(x_t)+\nabla x \log p(y x_t)$ Classifier guidance basis.	$\epsilon_{\text{guided}}=(1+w)\epsilon\theta(x_t,y,t)-w\epsilon\theta(x_t,\emptyset,t)$ Classifier-free guidance.

$FID=||\mu_r-\mu_g||^2+Tr(\Sigma_r+\Sigma_g-2(\Sigma_r\Sigma_g)^{\{1/2\}})$ Feature-distribution distance.

6. Equation symbol key table

This table explains the main symbols used in the equations. When studying, cover the meaning column and check whether you can explain each symbol clearly.

Symbol	Meaning	Plain-English reading
$s_t(x)$	Score	Gradient of log density at noise level t .
$p_t(x)$	Noisy data distribution	Distribution of samples at timestep t .

Symbol	Meaning	Plain-English reading
y	Condition	Class label, prompt, or target attribute.
w	Guidance scale	Strength of conditional push.
$\emptyset$	Null condition	Unconditional branch input.
μ_r, μ_g	Feature means	Average real/generated feature vectors.
Σ_r, Σ_g	Feature covariances	Spread/correlation of real/generated

features.

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7. Main code prompt / implementation

skeleton This is not meant to replace the lecture code. It is a compact prompt showing the computational pattern you should recognize in homework and exams.

Classifier-free guidance prompt

```
noise_uncond = model(xt, t, condition=None) noise_cond = model(xt, t, condition=y) noise_guided = noise_uncond + scale * (noise_cond - noise_uncond) xt_minus_1 = sampler_step(xt, t, noise_guided)
```

How to explain the code in words
Start by identifying the input, the model output, the objective being minimized, and the update rule. Then connect the code back to the formulas and the method table.

8. Common exam traps

Trap	Correct way to think
Using a clean classifier for	guidance Guidance queries noisy x_t , so the classifier should be noise-aware.
Assuming more guidance is	always better More guidance can reduce diversity and create artifacts.
Treating FID as absolute truth	FID is a feature proxy and can be misleading.

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9. Exam-style sample questions with

answers These are written in the style of conceptual exam answers. Use them to practice explaining, not only recognizing terms.

Question 1

What is classifier guidance based on?

Sample answer

Bayes rule: conditional score equals unconditional score plus classifier log-probability gradient.

Question 2

Why must the classifier be noise-aware?

Sample answer

The reverse chain queries it on noisy x_t , not clean images.

Question 3

What is the core idea of CFG?

Sample answer

Combine conditional and unconditional predictions from the same model.

Question 4

Why can stronger guidance reduce diversity?

Sample answer

It pushes samples into a narrower condition-consistent region.

Question 5

What does FID measure?

Sample answer

Distance between real and generated feature distributions, using means and covariances.

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10. Flashcards for rapid review

Use these for active recall. Cover the answer column and try to answer before checking.

Card 1

Q: What does the score point toward?

A: Higher probability regions.

Card 2

Q: What does CFG avoid?

A: A separate external classifier.

Card 3

Q: What does w control?

A: Guidance strength.

Card 4

Q: Why add stochasticity?

A: To preserve distributional sampling.

Card 5

Q: Is FID enough?

A: No, it is only a proxy.

Card 6

Q: What is noise-aware classification?

A: Classification trained on noisy x_t and t .

Card 7

Q: What happens if guidance is too strong?

A: Artifacts and reduced diversity.

Card 8

Q: What is drift?

A: Deterministic denoising movement.

Card 9

Q: What must you remember about ELBO to MSE?

A: DDPM noise MSE emerges from a variational objective under Gaussian assumptions.

Card 10

Q: What must you remember about Score?

A: $\nabla_x \log p_t(x)$ points toward higher density at a given noise level.

Card 11

Q: What must you remember about Reverse dynamics?

A: Sampling combines deterministic drift and stochastic noise.

Card 12

Q: What must you remember about Classifier guidance?

A: Use a noise-aware classifier gradient to bias samples toward y .

Card 13

Q: What must you remember about CFG?

A: Train conditional and unconditional branches, then combine predictions.

Card 14

Q: What must you remember about FID?

A: Compare real/generated feature means and covariances, but treat it as a proxy.

Card 15

Q: What does this formula represent: $s_t(x) = \nabla_x \log p_t(x)$?

A: Score at noise level t .

Card 16

Q: What does this formula represent:

$s_t(x_t) \approx -\epsilon \theta(x_t, t) / \sqrt{(1 - \bar{\alpha}_t)}$?

A: Noise prediction to score identity.

Card 17

Q: What does this formula represent: $\nabla_x \log p(x_t|y) = \nabla_x \log p(x_t) + \nabla_x \log p(y|x_t)$?

A: Classifier guidance basis.

Card 18

Q: What does this formula represent:

$$\epsilon_{\text{guided}} = (1+w)\epsilon_{\theta}(x_t, y, t) - w\epsilon_{\theta}(x_t, \emptyset, t)?$$

A: Classifier-free guidance.

Card 19

Q: What does this formula represent:

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})?$$

A: Feature-distribution distance.

Card 20

Q: What does $s_t(x)$ mean?

A: Score. Gradient of log density at noise level t .

Card 21

Q: What does $p_t(x)$ mean?

A: Noisy data distribution. Distribution of samples at timestep t .

Card 22

Q: What does y mean?

A: Condition. Class label, prompt, or target attribute.

Card 23

Q: What does w mean?

A: Guidance scale. Strength of conditional push.

Card 24

Q: What does $\emptyset$ mean?

A: Null condition. Unconditional branch input.

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Card 25

Q: What does μ_r, μ_g mean?

A: Feature means. Average real/generated feature vectors.

Card 26

Q: What does Σ_r, Σ_g mean?

A: Feature covariances. Spread/correlation of real/generated features.

Card 27

Q: What is the key idea of Why MSE appears?

A: The simple noise-prediction loss comes from the probabilistic DDPM structure. With Gaussian forward conditionals and fixed/appropriate reverse variance, the intermediate KL terms reduce to errors in the reverse mean,.

Card 28

Q: What is the key idea of Score perspective?

A: The score $\nabla_x \log p_t(x)$ points in the direction where the noisy-data distribution becomes more likely. For Gaussian-corrupted data, predicting noise is closely tied to predicting this

Card 29

Q: What is the key idea of Reverse drift plus stochasticity?

A: The deterministic drift moves samples toward high-probability data regions. The stochastic term preserves sampling diversity and prevents the process from becoming pure mode-seeking gradient

Card 30

Q: What is the key idea of Classifier guidance?

A: Bayes rule shows that a conditional score can be decomposed into an unconditional score plus a classifier gradient. The diffusion model provides realism; the classifier provides condition

Card 31

Q: What is the key idea of Classifier-free guidance?

A: CFG trains one model with the condition sometimes present and sometimes dropped. At sampling time, conditional and unconditional predictions are combined to strengthen condition

Card 32

Q: What is the key idea of Evaluation and FID?

A: FID compares real and generated feature distributions using means and covariances. It is useful but not absolute; it can miss prompt correctness, diversity problems, and domain

Card 33

Q: What happens in the step: Learn score-related field?

A: Noise prediction can be interpreted as learning a denoising/score direction.

Card 34

Q: What happens in the step: Add condition?

A: Use classifier guidance or CFG to bias generation toward y or text.

Card 35

Q: What happens in the step: Tune guidance scale?

A: Increase adherence carefully because diversity may fall.

Card 36

Q: What happens in the step: Evaluate critically?

A: Use FID plus visual and semantic checks.

Card 37

Q: Can you define score?

A: Gradient of log density.

Card 38

Q: Can you explain CFG?

A: Amplify conditional prediction against unconditional baseline.

Card 39

Q: Can you explain guidance tradeoff?

A: Adherence vs diversity/artifacts.

Card 40

Q: Can you explain FID limits?

A: Feature proxy, domain/sample-size sensitive.

Card 41

Q: What is classifier guidance based on?

A: Bayes rule: conditional score equals unconditional score plus classifier log-probability gradient.

Card 42

Q: Why must the classifier be noise-aware?

A: The reverse chain queries it on noisy x_t , not clean images.

Card 43

Q: What is the core idea of CFG?

A: Combine conditional and unconditional predictions from the same model.

Card 44

Q: Why can stronger guidance reduce diversity?

A: It pushes samples into a narrower condition-consistent region.

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Card 45

Q: What does FID measure?

A: Distance between real and generated feature distributions, using means and covariances.

Card 46

Q: In plain English, why is $s_t(x) = \nabla_x \log p_t(x)$ important?

A: It captures: Score at noise level t . Use it as a compact exam sentence, then define each symbol.

Card 47

Q: In plain English, why is $s_t(x_t) \approx -\epsilon\theta(x_t, t) / \sqrt{(1 - \bar{\alpha}_t)}$ important?

A: It captures: Noise prediction to score identity.

Use it as a compact exam sentence, then define each symbol.

Card 48

Q: In plain English, why is $\nabla_x \log p(x_t|y) = \nabla_x \log p(x_t) + \nabla_x \log p(y|x_t)$ important?

A: It captures: Classifier guidance basis. Use it as a compact exam sentence, then define each symbol.

Card 49

Q: In plain English, why is $\epsilon_{\text{guided}} = (1+w)\epsilon\theta(x_t, y, t) - w\epsilon\theta(x_t, \emptyset, t)$ important?

A: It captures: Classifier-free guidance. Use it as a compact exam sentence, then define each symbol.

Card 50

Q: In plain English, why is $\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$ important?

A: It captures: Feature-distribution distance. Use it as a compact exam sentence, then define each symbol.

Card 51

Q: In plain English, why is $s_t(x) = \nabla_x \log p_t(x)$ important?

A: It captures: Score at noise level t . Use it as a compact exam sentence, then define each symbol.

Card 52

Q: In plain English, why is $s_t(x_t) \approx -\epsilon\theta(x_t, t) / \sqrt{(1 - \bar{\alpha}_t)}$ important?

A: It captures: Noise prediction to score identity.

Use it as a compact exam sentence, then define each symbol.

Card 53

Q: In plain English, why is $\nabla_x \log p(x_t|y) = \nabla_x \log p(x_t) + \nabla_x \log p(y|x_t)$ important?

A: It captures: Classifier guidance basis. Use it as a compact exam sentence, then define each symbol.

Card 54

Q: In plain English, why is $\epsilon_{\text{guided}} = (1+w)\epsilon\theta(x_t, y, t) - w\epsilon\theta(x_t, \emptyset, t)$ important?

A: It captures: Classifier-free guidance. Use it as a compact exam sentence, then define each symbol.

Card 55

Q: In plain English, why is $\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$ important?

A: It captures: Feature-distribution distance. Use it as a compact exam sentence, then define each symbol.

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Explanation

Chapter 4: Advanced Diffusion Concepts

How this section is organized

Each item below is a heading taken from the corresponding Jupyter notebook in the same order it appears there. The explanation beside it tells you what that notebook section is trying to teach and why it matters for the review.

Source notebook

Chapter_4_Adv_Diffusion(1).ipynb

Headings included

25

Notebook headings with explanation

1. Chapter 4 - Advanced Concepts in Diffusion

After this chapter you should be able to: 1. Understand that error minimization in reverse diffusion is related to ELBO 2.

2. Error Estimation and the ELBO in Diffusion Models

The Probabilistic Objective Diffusion models are trained by maximizing the log-likelihood: $\log p_{\theta}(x_0)$ Since the marginal likelihood is intractable, we optimize a variational lower bound (ELBO): $\log p_{\theta}(x_0) \geq L_{ELBO}$ Forward and Reverse Processes Forward diffusion (fixed, known): $q(x_t | x_{t-1}) = N(x_t | \sqrt{\alpha_t} x_{t-1}, \beta_t)$ Reverse model (learned): $p_{\theta}(x_{t-1} | x_t) = N(x_{t-1} | \mu_{\theta}(x_t, t), \Sigma_{\theta}(x_t, t))$ Full ELBO Expression The complete ELBO for.

3. What is a Score?

Definition: $s(x) = d/dx \log p(x)$ It is simply the slope of the log-density curve. It tells us: - Which direction increases probability - How strongly probability increases Example 1: Gaussian $N(0,1)$ For $p(x) = N(\mu, \sigma^2)$, using the definition we can derive the log. $\log p(x) = -C - (x-\mu)^2 / (2\sigma^2)$ is a downward parabola with its maximum at $x=\mu$. The notebook also includes 2 nearby code cells for implementation, demonstration, or visualization of this heading.

4. Review: Forward Diffusion

Closed-form noising: $x_t = \sqrt{\alpha_t} x_0 + \sqrt{1-\alpha_t} \epsilon$, $\epsilon \sim N(0,1)$. The network is trained to predict: $\epsilon_{\theta}(x_t, t) \approx \epsilon$.

5. Reverse Diffusion = Score Step + Noise

Reverse update can now be written directly in score form: $x_{t-1} = x_t + \beta_t s_t(x_t) + \sqrt{\beta_t} z$, $z \sim N(0,1)$. Substitute the score expression: $x_{t-1} = x_t - \beta_t / \sqrt{1-\alpha_t} \epsilon_{\theta}(x_t, t) + \sqrt{\beta_t} z$.

6. Example: 1D Reverse Simulation (Score + Noise Only)

This heading introduces a practical example or code activity. Its purpose is to connect the theory to an executable workflow or visualization. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.

7. Full MNIST Example (Score / Noise-Prediction Diffusion)

This section shows a complete diffusion model on MNIST using the score approach. Key points: - We train a neural network to predict the forward noise ϵ : $\epsilon_{\theta}(x_t, t) \approx \epsilon$. - That prediction gives a score estimate: $s_t(x_t) \approx -1/\sqrt{1-\alpha_t} \epsilon_{\theta}(x_t, t)$. - Sampling is “subtract predicted noise + add calibrated Gaussian noise” over timesteps. The notebook also includes 2 nearby code cells for implementation, demonstration, or visualization of this heading.

8. Inject time via FiLM-like bias

`self.to_bias1 = nn.Linear(base*4, base)` `self.to_bias2 = nn.Linear(base*4, base*2)` `self.to_bias3 = nn.Linear(base*4, base*4)` What Problem Is This Solving? In diffusion models, the neural network must depend on time step t : $\epsilon_{\theta}(x_t, t)$ The network must behave differently at early timesteps (very noisy inputs) and late timesteps (almost clean images).

9. Important point to note	The original convolution bias b : - is constant across all timesteps. The time bias b_t : - changes for every timestep. - is computed dynamically during the forward pass.
10. Summary	These lines implement: Time Embedding -> Per-layer Bias which allows the network to: - Know how noisy the input is - Adjust its denoising behavior accordingly - Learn different score fields at different timesteps Without time injection, diffusion models would fail. Diffusion models are not just image-to-image networks. The notebook also includes 3 nearby code cells for implementation, demonstration, or visualization of this heading.
11. Reverse DDPM Step — Code ↔ Math (VSCode-Safe)	This heading introduces a practical example or code activity. Its purpose is to connect the theory to an executable workflow or visualization.
12. Guided Diffusion	The Limitation of Unconditional Diffusion A standard diffusion model learns the data distribution: $p_\theta(x)$ It is trained to reverse the forward noise process: $x_t = \sqrt{\alpha_t} x_0 + \sqrt{1-\alpha_t} \epsilon$ and learns a score field (via noise prediction): $\nabla_x \log p_\theta(x_t)$ During sampling, it produces samples from the learned data distribution: $x_{t-1} = \mu_\theta(x_t, t) + \sqrt{\beta_t} z$ or equivalently (score form): $x_{t-1} = x_t + \beta_t s_\theta(x_t, t) + \sqrt{\beta_t} z$ This works extremely well.
13. Two Main Ways to Guide Diffusion	Classifier Guidance Modify score: $s_t^{\text{guided}}(x) = s_t(x) + w \nabla_x \log p_\phi(y \mid x, t)$ Interpretation: - The diffusion score ensures realism. - The classifier gradient ensures condition alignment. Classifier-Free Guidance (CFG) Combine conditional and unconditional predictions: $\text{varepsilon}_{\text{cfg}} = (1+w)\text{varepsilon}_\theta(x_t, t, y) - w\text{varepsilon}_\theta(x_t, t, \text{varnothing})$ Interpretation: - Amplifies what distinguishes conditional from unconditional predictions. - No external classifier required.
14. Step-by-step methodology to build guided diffusion (both approaches)	This is a practical checklist you can follow to build guided diffusion in two mainstream ways. Approach A — Classifier guidance (external classifier $p_\phi(y \mid x, t)$) Goal: sample from $p(x \mid y)$ using an unconditional diffusion model plus an auxiliary classifier.
15. Guided Diffusion (1D Examples)	Guided diffusion modifies the direction field used in the reverse step. Recall the score-form reverse step: $x_{t-1} = x_t + \beta_t s_\theta(x_t, t) + \sqrt{\beta_t} z$; $z \sim \mathcal{N}(0, 1)$. The notebook also includes 2 nearby code cells for implementation, demonstration, or visualization of this heading.
16. MNIST: Classifier Guidance vs. Classifier-Free Guidance + FID-like Comparison	Train a small MNIST classifier (also used as a feature extractor). Use classifier guidance with the unconditional diffusion model. The notebook also includes 4 nearby code cells for implementation, demonstration, or visualization of this heading.
17. Reverse Diffusion Sampling Using Score — Code ↔ Math (Line-by-Line)	We now explain the reverse diffusion sampling loop and connect each part to the DDPM equations. Initialize from Pure Noise Code $x = \text{torch.randn}(n, 1, 28, 28, \text{device=device})$ Math We start from the prior: $x_T \sim \mathcal{N}(0, I)$ This is the terminal distribution of the forward process: $q(x_T \mid x_0) \approx \mathcal{N}(0, I)$ So generation begins at pure noise. The notebook also includes 2 nearby code cells for implementation, demonstration, or visualization of this heading.
18. For Classifier Guided MNIST we need to train a classifier first	This notebook heading introduces for classifier guided mnist we need to train a classifier first and explains how it supports the chapter concepts, equations, method steps, or implementation workflow. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.

19. Fréchet Inception Distance (FID)	The Fréchet Inception Distance (FID) is a metric used to evaluate the quality of images generated by a generative model (GANs, diffusion models, etc.). It measures how close the distribution of generated images is to the distribution of real images. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.
20. Side Note: Explanation for model.eval() in the code	Why Do We Need model.eval() and clf.eval()? When sampling from a trained diffusion model (and classifier), we use: model.eval() clf.eval() This switches the networks into evaluation mode.
21. Classifier-Guided Diffusion — Code ↔ Math (Side-by-Side)	We now explain classifier-guided reverse diffusion line by line. From conditional identity: $\nabla_x \log p(x \mid y) = \nabla_x \log p(x) + \nabla_x \log p(y \mid x)$ We are approximating the second term.
22. MNIST Classifier-Guided Sampling	This notebook heading introduces mnist classifier-guided sampling and explains how it supports the chapter concepts, equations, method steps, or implementation workflow. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.
23. Classifier-Free Guidance (CFG) Training — Code ↔ Math for MNIST	We now explain how classifier-free guidance is trained and connect each line of code to the mathematical objective. Forward Diffusion (Closed Form) Code $x_t, \epsilon = q_sample(x_0, t)$ Math Sample from: $x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon$ where: $\epsilon \sim N(0, I)$ The training target is ϵ .
24. Classifier-Free Guidance (CFG) Sampling — Code ↔ Math for MNIST Example	We now explain how CFG modifies reverse diffusion at sampling time. Switch to Evaluation Mode Code <code>cfg_model.eval()</code> Math Freeze dropout / BatchNorm behavior. The notebook also includes 2 nearby code cells for implementation, demonstration, or visualization of this heading.
25. Interpreting the MNIST-FID Results (for Digit 3)	We evaluated conditional generation for digit 3 using FID computed in classifier feature space. Results: Unconditional: 327.33 Classifier guidance: 41.76 Classifier-free guidance: -31.17 Unconditional FID = 327.33 This is very high.